

Naphtha Hydrodesulfurization

Process Simulation, Heat Integration & Techno-Economic Analysis

Aspen Plus V12 | Aspen Energy Analyzer | APEA

Simulation Tool:	Aspen Plus V12
Thermodynamic Model:	Peng-Robinson EOS
Feed:	3000 kg/hr Naphtha (C ₅ –C ₁₂)
Reactor Conditions:	T = 350°C, P = 40 bar (Isothermal)
Catalyst:	Co-Mo / γ -Al ₂ O ₃
Course:	Petroleum Refining & Process Design
Date:	April 2026

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Chapter 1

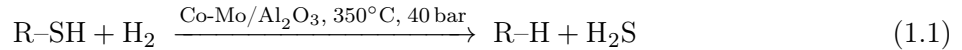
Introduction & Background Theory

1.1 Overview and Environmental Motivation

Naphtha (boiling range 30–200°C, C₅–C₁₂) is the primary feedstock for catalytic reforming and petrochemicals, but requires desulfurization before the reformer because even trace sulfur permanently poisons Pt/Re catalysts. Global fuel regulations now mandate ≤ 10 ppm sulfur, driving massive HDS investment. India’s BS-VI mandate (2020) required all major refineries (HPCL, BPCL, IOC) to upgrade HDS units. HDS processes collectively treat 60–70% of all refinery streams worldwide.

1.2 HDS Reaction Chemistry and Kinetics

HDS reacts sulfur compounds with H₂ over a Co-Mo/Al₂O₃ catalyst:



All HDS reactions are exothermic ($\Delta H^\circ \approx -66$ to -280 kJ/mol) and thermodynamically favorable ($>99.9\%$ equilibrium conversion at 300°C, 40 bar). Rate follows Langmuir-Hinshelwood kinetics:

$$r = \frac{k \cdot P_{\text{H}_2}^m \cdot C_S}{(1 + K_{\text{H}_2\text{S}} \cdot P_{\text{H}_2\text{S}})^n} \quad (1.2)$$

H₂S product inhibits the effective rate by 20–40%. Adiabatic temperature rise is 15–40°C for naphtha feeds.

Table 1.1: Sulfur Compounds, Reactions and Relative Reactivity

Compound	Reaction	Reactivity	Conv.
Methanethiol (CH ₄ S)	CH ₄ S + H ₂ → CH ₄ + H ₂ S	Fast	~100%
Ethanethiol (C ₂ H ₆ S)	C ₂ H ₆ S + H ₂ → C ₂ H ₆ + H ₂ S	Fast	~100%
1-Butanethiol (C ₄ H ₁₀ S)	C ₄ H ₁₀ S + H ₂ → C ₄ H ₁₀ + H ₂ S	Moderate	~100%
Dibenzothiophene (C ₁₂ H ₈ S)	C ₁₂ H ₈ S + 3H ₂ → C ₁₂ H ₁₂ + H ₂ S	Slow	>95%

1.3 Catalyst Selection: Co-Mo vs Ni-Mo

Table 1.2: Comparison of Co-Mo and Ni-Mo HDS Catalysts

Property	Co-Mo/Al ₂ O ₃ (this work)	Ni-Mo/Al ₂ O ₃
Best for	Naphtha, light feeds (DDS route)	Heavy gas oil, diesel, DBT (HYD)
Temperature	280–340°C	320–380°C
Pressure	20–50 bar H ₂	40–80 bar H ₂
H ₂ use	Lower	20–30% higher
Catalyst life	2–4 years (regenerable)	2–4 years
LHSV	1–4 hr ⁻¹	0.5–2 hr ⁻¹

Co-Mo was selected for this naphtha HDS simulation due to its superior performance on light, thiol-rich feeds at lower pressures via the Direct Desulfurization (DDS) pathway.

Chapter 2

Process Description & Aspen Plus Simulation

2.1 Main Process Flowsheet

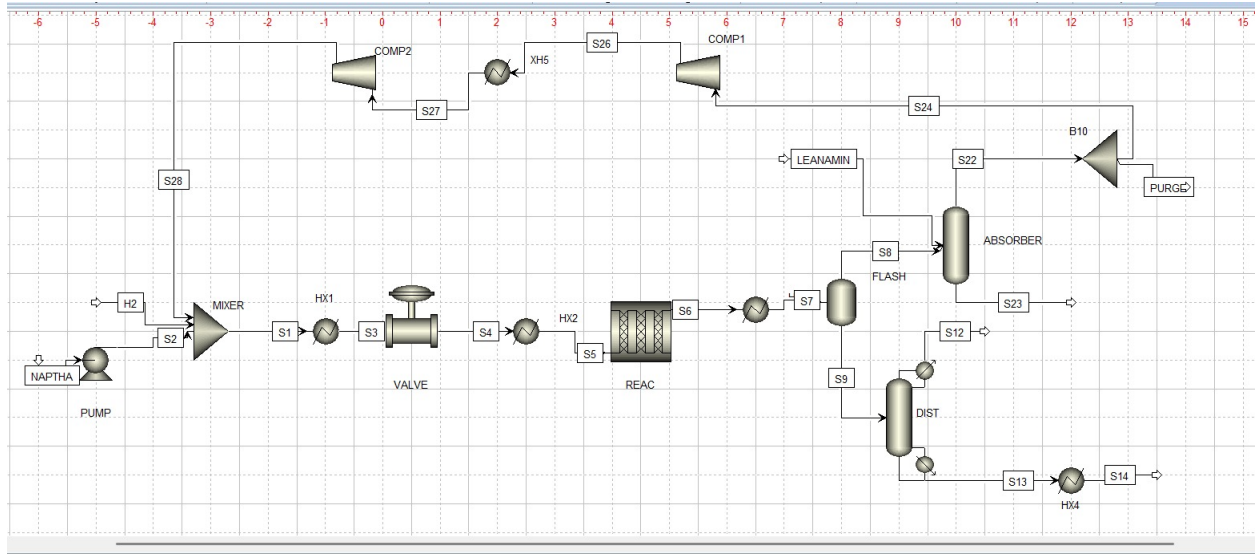


Figure 2.1: Aspen Plus Process Flowsheet: Naphtha HDS showing PUMP, MIXER, HX1, VALVE, HX2, REAC, HX3, FLASH, ABSORBER, DIST, COMP1, COMP2, B10 splitter, and streams S1–S28.

2.2 Process Flow Description

- Feed Preparation:** Naphtha (3000 kg/hr, 25°C, 2 bar) pressurized to 41 bar by PUMP; mixed with fresh H₂ (4.43 kmol/hr) and recycle H₂ (S28, 7.04 kmol/hr) in MIXER.
- Preheating:** Mixed stream preheated in HX1 (38.8→250°C via process-to-process heat exchange), throttled through VALVE (41→40 bar), then heated to 350°C in HX2.
- Reaction:** REAC (RStoic, 350°C, 40 bar, isothermal) achieves 100% conversion of all sulfur species to H₂S.
- Flash Separation:** Reactor effluent cooled in HX3 and fed to FLASH (35°C, 4 bar) separating vapor (H₂, H₂S, light HC) from liquid naphtha.
- H₂S Removal:** Flash vapor (S8) enters ABSORBER with 50 kmol/hr lean amine. H₂S-free H₂ overhead (S22) is split: 10% purged (B10), 90% recompressed (COMP2) and recycled.
- Product Recovery:** Flash liquid fed to DIST (RadFrac) separating light HC (S12) from desulfurized naphtha bottoms (S13→HX4→S14, 2526 kg/hr at 40°C).

2.3 Key Stream Data and Material Balance

Table 2.1: Overall Material Balance — Key Process Streams

Parameter	Feed/Inlet	Product/Outlet	Significance
Naphtha mass flow	3000 kg/hr	2526 kg/hr (S14)	S and light HC removed
Total sulfur	~259.5 kg/hr	≈0 kg/hr	100% desulfurization
H ₂ S produced	—	81.19 kg/hr	Sent to Claus plant
H ₂ fresh feed	4.43 kmol/hr	—	Limiting reactant
H ₂ recycle (S28)	7.04 kmol/hr	—	70–85% H ₂ saving
PURGE stream	—	11.38 kg/hr (MW=14.55)	Mostly CH ₄
Reactor T/P	350°C / 40 bar	Isothermal	Design basis

2.4 Equipment Blocks and Property Method

The Peng-Robinson EOS was used:

$$P = \frac{RT}{V - b} - \frac{a(T)}{V(V + b) + b(V - b)} \quad (2.1)$$

PR EOS Justification: Handles H₂/HC/H₂S at 40 bar (VLE accuracy $\pm 2-5\%$). Supercritical H₂ ($T_c=33$ K) handled via Boston-Mathias extrapolation. Binary k_{ij} for CH₄/H₂ and H₂/H₂S from Aspen databank. Tear stream S28 converged via Wegstein method ($\alpha=0.5$, 50 iterations, tolerance 0.01% on mole flows). REAC modeled as RStoic (100% conversion, isothermal). ABSORBER and DIST as RadFrac.

Chapter 3

Heat Integration & Pinch Analysis

3.1 Aspen Energy Analyzer — Savings Summary

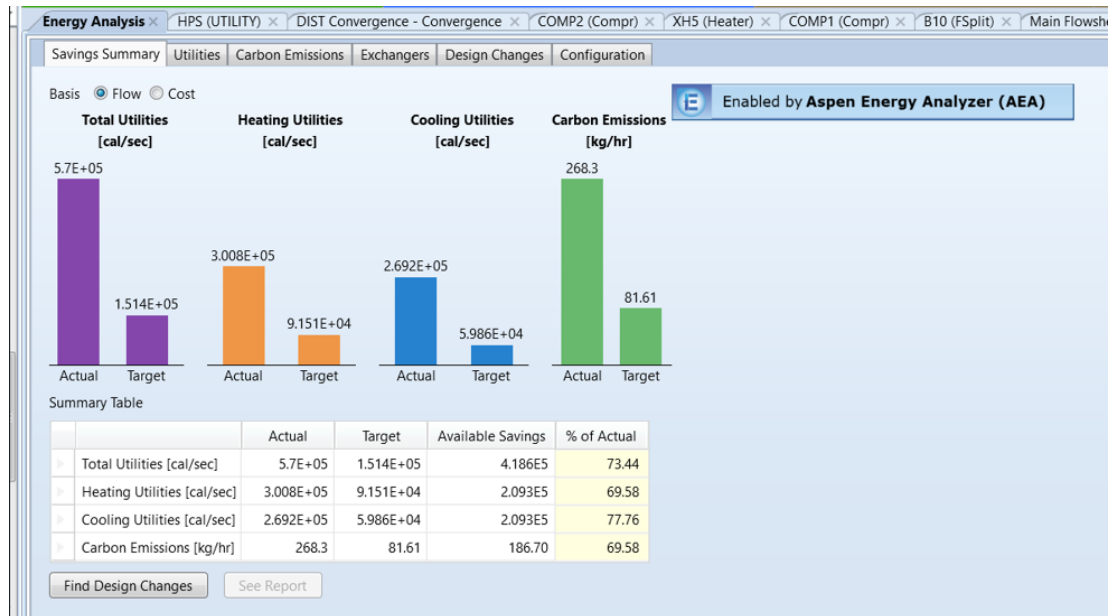


Figure 3.1: AEA Savings Summary (Enabled by Aspen Energy Analyzer): Bar charts comparing Actual vs. Target for Total Utilities ($5.7\text{E}+05$ vs $1.514\text{E}+05$ cal/sec), Heating Utilities ($3.008\text{E}+05$ vs $9.151\text{E}+04$ cal/sec), Cooling Utilities ($2.692\text{E}+05$ vs $5.986\text{E}+04$ cal/sec), and Carbon Emissions (268.3 vs 81.61 kg/hr). Total savings potential = **73.44%**.

3.2 Detailed Utility Analysis (AEA Utilities Tab)

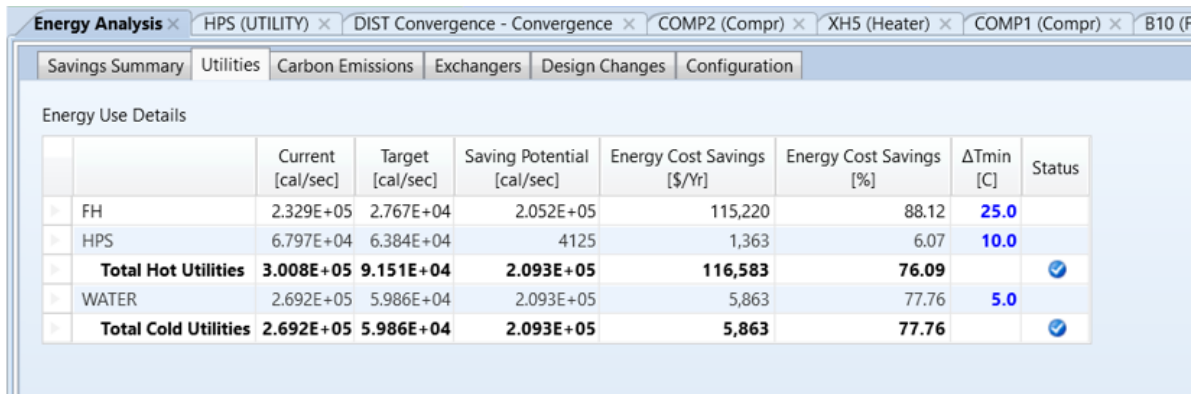


Figure 3.2: AEA Energy Use Details (Utilities Tab): Individual utility breakdown for FH (Furnace, $\Delta T_{min}=25^{\circ}\text{C}$), HPS (High-Pressure Steam, $\Delta T_{min}=10^{\circ}\text{C}$), and WATER (cooling, $\Delta T_{min}=5^{\circ}\text{C}$). Shows Current, Target, Saving Potential (cal/sec), Energy Cost Savings (\$/yr and %) per utility.

3.3 Carbon Emissions Analysis (AEA Carbon Emissions Tab)

Energy Analysis × HPS (UTILITY) × DIST Convergence - Convergence × COMP2 (Compr) × XH5 (Heater) × CC					
Savings Summary Utilities Carbon Emissions Exchangers Design Changes Configuration					
Carbon Emissions Details					
	Current [kg/hr]	Target [kg/hr]	Saving Potential [kg/hr]	Emission Cost Savings [\$/Yr]	Emission Cost Savings [%]
FH	207.7	24.68	183	0	0.00
HPS	60.61	56.93	3.679	0	0.00
Total Hot Utilities	268.3	81.61	186.7	0	0.00
WATER	0	0	0	0	0.00
Total Cold Utilities	0	0	0	0	0.00

Figure 3.3: AEA Carbon Emissions Details: Current vs. Target CO₂-equivalent emissions (kg/hr) for FH (207.7→24.68 kg/hr) and HPS (60.61→56.93 kg/hr). Total saving potential = **186.7 kg/hr** (69.58% reduction).

A 186.7 kg/hr CO₂ reduction (69.58%) from the fired heater is equivalent to \$0.4–0.8M/yr benefit at a carbon price of \$50/tonne CO₂ for a full industrial-scale unit.

3.4 Composite Curves and Pinch Point

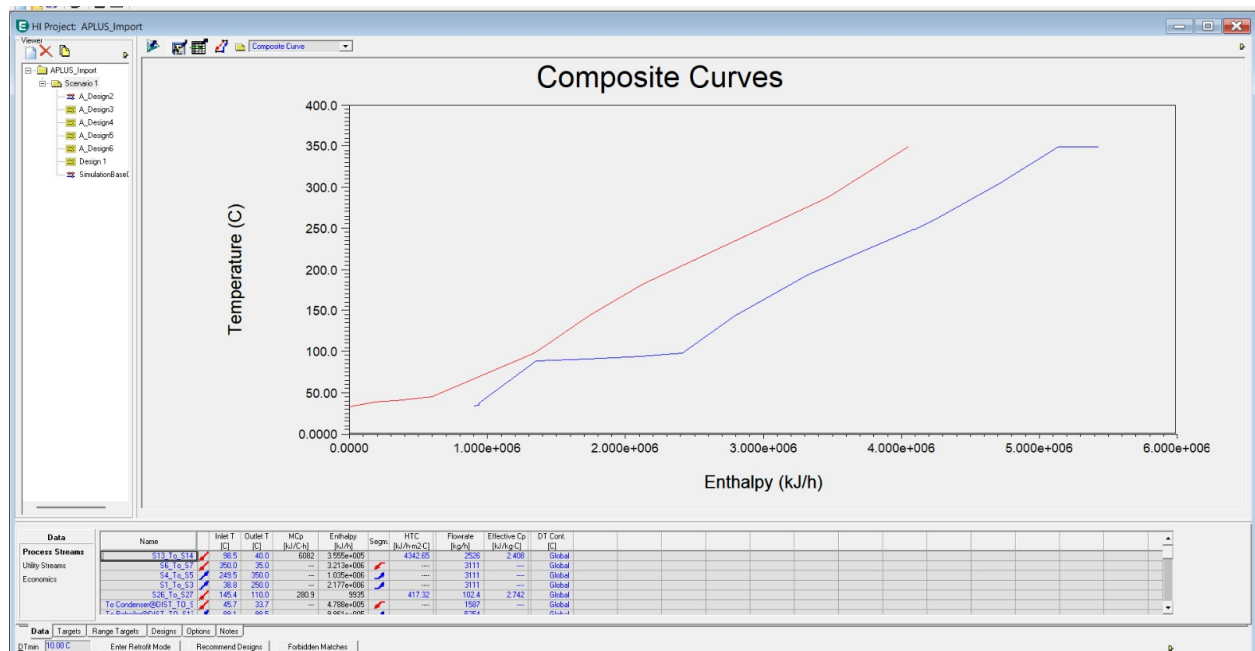


Figure 3.4: Composite Curves from AEA (HI Project: APLUS_Import, Scenario 1). **Red curve** = Hot Composite (process streams to be cooled, max ≈ 350°C — reactor outlet). **Blue curve** = Cold Composite (streams to be heated; plateau at 90–100°C corresponds to DIST reboiler). $\Delta T_{min} = 10^\circ\text{C}$. The overlap region represents maximum recoverable heat (process-to-process transfer). Lower panel shows stream data: S13→S14, S6→S7, S4→S5, S2→S3, S26→S27, DIST condenser and reboiler.

Pinch Analysis Key Results:

Pinch point near 40–50°C (cold composite). **Dominant heat recovery match:** hot reactor effluent (S6→S7: 350→35°C) preheats cold mixed feed (S2→S3: 38.8→250°C) via HX1 — this single match delivers the majority of heat recovery. **Min. hot utility** $Q_{H,min} = 9.151 \times 10^4$ cal/sec; **Min. cold utility** $Q_{C,min} = 5.986 \times 10^4$ cal/sec. AEA synthesized 6 candidate HEN designs (A_Design2 to A_Design6) under Scenario 1.

Table 3.1: Process Streams for HEN Synthesis (AEA Data Panel)

Stream	T_{in} (°C)	T_{out} (°C)	MCp (kJ/h°C)	Enthalpy (kJ/h)	Type
S13→S14	98.5	40.0	6082	3.555×10^5	Hot
S6→S7 (REAC out)	350.0	35.0	—	3.213×10^6	Hot
S4→S5	249.5	350.0	—	1.035×10^6	Cold
S2→S3 (feed)	38.8	250.0	—	2.177×10^6	Cold
S26→S27	145.4	110.0	9935	—	Hot
DIST Condenser	45.7	33.7	417	4.788×10^5	Hot
DIST Reboiler	99.1	99.5	6354	—	Cold

Chapter 4

Sensitivity Analysis

Four sensitivity studies were conducted in Aspen Plus (Sensitivity block S-1) varying one key parameter at a time while monitoring critical process outputs.

4.1 Sensitivity 1: H₂ Feed Flow Rate vs. Reactor Performance

Varied: H₂ molar flow (MIXED), 2.0–8.0 kmol/hr **Responses:** H₂S mole fraction at S6; Product molar flow (PRO KMOL/HR)

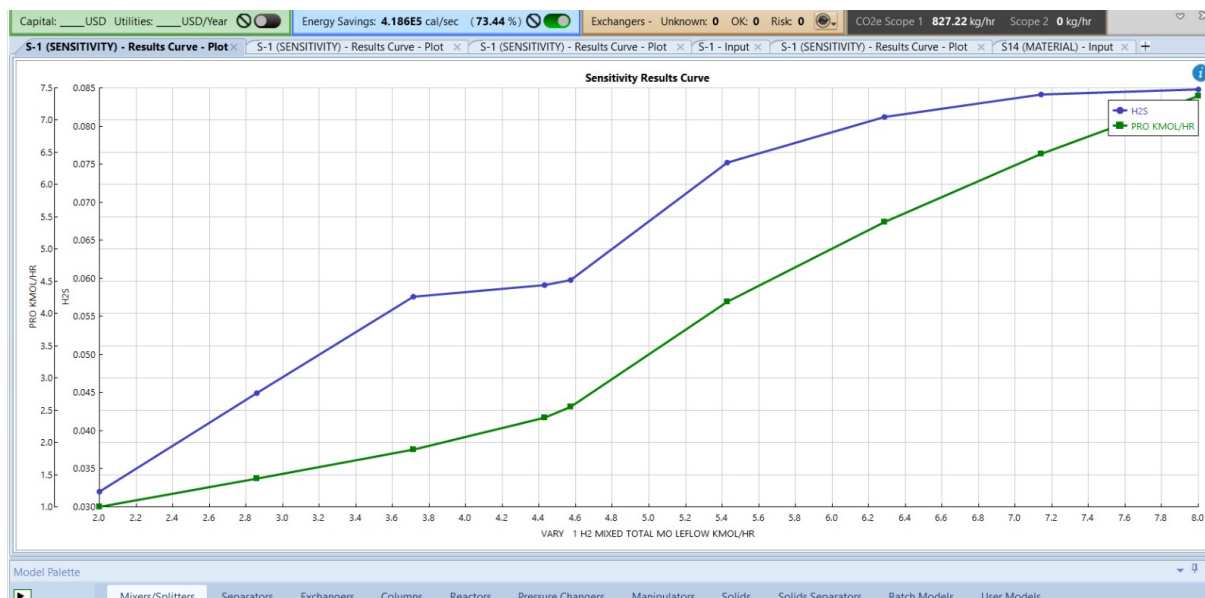


Figure 4.1: Sensitivity 1 — H₂ Feed Flow Rate vs. Reactor Performance. **Blue curve (left axis, H₂S):** H₂S mole fraction at reactor outlet S6 (extent of desulfurization). **Green curve (right axis, PRO KMOL/HR):** Product molar flow rate. Both exhibit nonlinear S-shaped behavior: steep rise below 4.43 kmol/hr (H₂ limiting), gradual plateau above 5.4 kmol/hr (sulfur limiting). Design point at 4.43 kmol/hr sits at the inflection region.

Table 4.1: Sensitivity 1 Results

H ₂ (kmol/hr)	H ₂ S MF	PRO (kmol/hr)
2.0	0.031	1.00
2.8	0.046	1.35
3.7	0.058	1.88
4.43 (design)	0.060	2.43
5.4	0.075	4.57
7.2	0.084	6.57
8.0	0.085	7.42

Conclusion: H₂ is limiting below 4.43 kmol/hr. Beyond 6 kmol/hr, diminishing returns confirm sulfur compounds as the new limiting species. **Optimal: 4.43–5 kmol/hr** — the inflection point gives best balance of H₂ consumption vs. sulfur conversion.

4.2 Sensitivity 2: Flash Drum Pressure vs. H₂ Recovery and H₂S Separation

Varied: Flash drum pressure (FLASH PARAM PRES), 3–20 bar **Responses:** H₂S mole frac. in vapor S8 (S8H₂S); H₂ mole frac. in S8 (H₂REC)

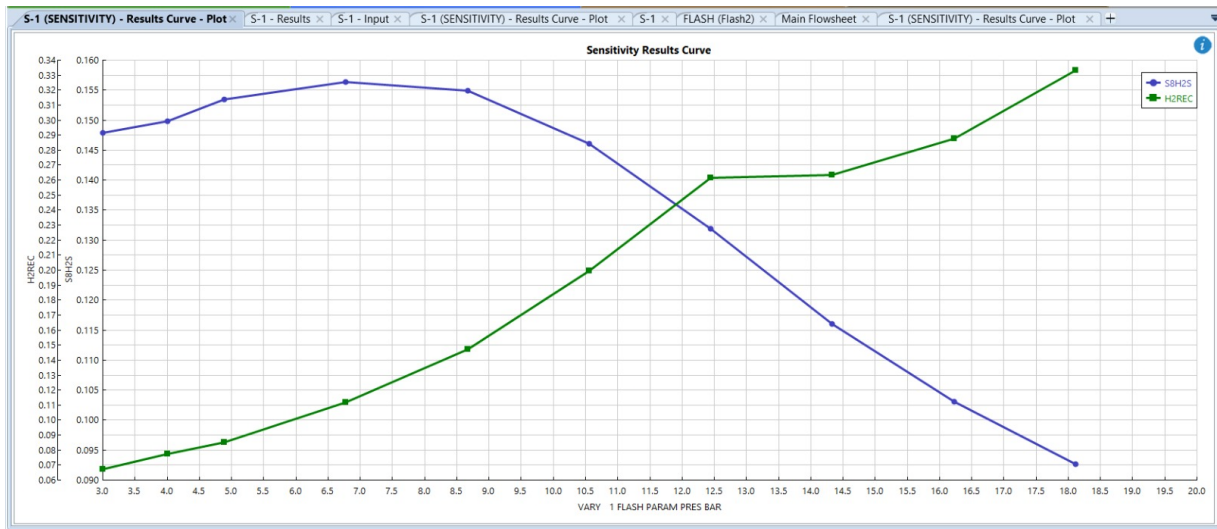


Figure 4.2: Sensitivity 2 — Flash Drum Pressure vs. Vapor Phase Composition. **Blue curve (left axis, S8H2S):** H_2S mole fraction in flash vapor — peaks at 0.156 near 7 bar then decreases sharply as higher pressure forces H_2S into liquid phase. **Green curve (right axis, H2REC):** H_2 mole fraction in vapor — increases monotonically from 0.091 (3 bar) to 0.158 (18 bar). Crossover at ≈ 12.5 bar = optimal flash pressure.

Table 4.2: Sensitivity 2 Results

P (bar)	S8H2S (MF)	H2REC (MF)
3.0	0.149	0.091
4.9	0.155	0.098
6.9	0.156 (max)	0.111
10.5	0.120	0.125
12.5 (cross)	0.116	0.140
16.5	0.113	0.140
18.0	0.072	0.158

Conclusion: Optimal flash pressure = 12–13 bar. Below the crossover, excessive H_2S carries over to the absorber. Above it, H_2 recovery improves while H_2S condenses into liquid, reducing absorber load.

4.3 Sensitivity 3: Lean Amine Flow Rate vs. H_2S Removal and H_2 Purity

Varied: LEANAMIN molar flow, 30–100 kmol/hr **Responses:** H_2S in purge (H2SPURGE, left axis); H_2 in recycle S24 (H224, right axis)

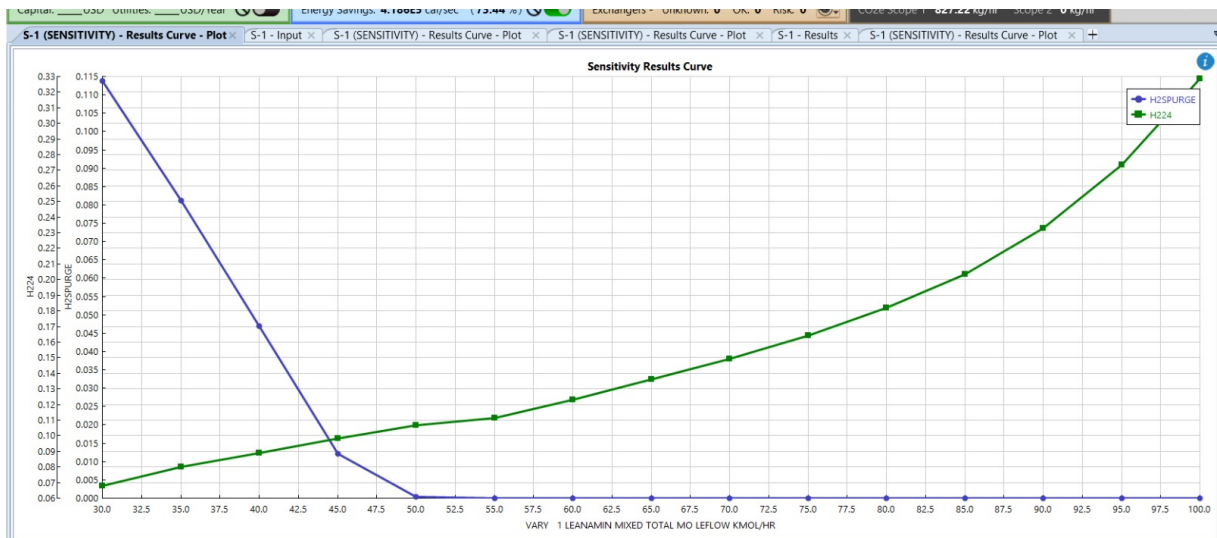


Figure 4.3: Sensitivity 3 — Lean Amine Flow vs. H_2S Removal and H_2 Purity. **Blue curve (left axis, H2SPURGE):** H_2S mole fraction in purge — drops steeply from 0.331 at 30 kmol/hr to zero at 50 kmol/hr. **Green curve (right axis, H224):** H_2 mole fraction in recycle S24 — increases from 0.005 to 0.32, confirming amine flow enriches H_2 recycle purity.

Table 4.3: Sensitivity 3 Results

Amine (kmol/hr)	H2SPURGE	H224 (MF)
30.0	0.331	0.005
35.0	0.252	0.008
40.0	0.165	0.014
45.0	0.093	0.015
50.0 (design)	0.000	0.020
80.0	0.000	0.053
100.0	0.000	0.320

Conclusion: Optimal lean amine = 50 kmol/hr (minimum for complete H₂S removal in purge). Higher amine simultaneously enriches H₂ recycle purity. The current design at 50 kmol/hr is at the minimum effective point, optimizing both cost and performance.

4.4 Sensitivity 4: Purge Fraction vs. H₂ Loss and Compressor Duty

Varied: B10 purge fraction, 0.05–0.50 **Responses:** H₂ loss (H2LOSS kmol/hr, blue), H₂ recycle (H2RE kmol/hr, green), Compressor duty (COMPDUTY kW, pink)

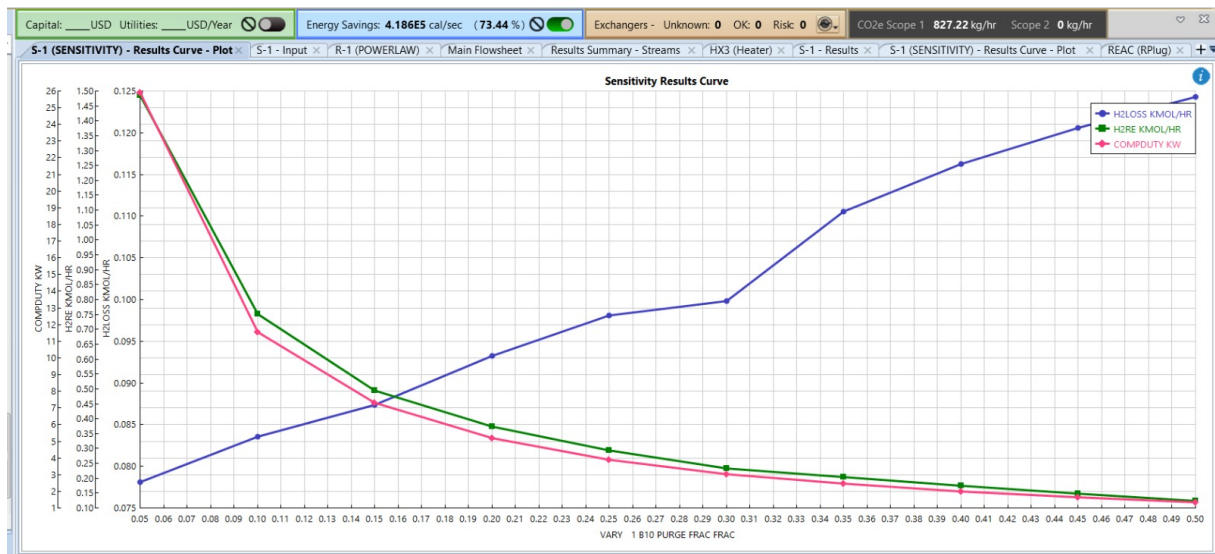


Figure 4.4: Sensitivity 4 — Purge Fraction vs. H₂ Loss, Recycle, and Compressor Duty. **Blue (H2LOSS):** increases with purge fraction. **Green (H2RE):** decreases as more is purged. **Pink (COMPDUTY kW, right axis):** decreases with higher purge (less recycle flow to compress). Crossover of H2LOSS and H2RE at purge fraction ≈ 0.15 = optimal trade-off.

Table 4.4: Sensitivity 4 Results

Purge	H2LOSS	H2RE
0.05	2.0	25.0
0.10	3.0	13.0
0.15	7.0	8.0
0.20	10.0	6.0
0.30	14.5	3.5
0.50	25.0	2.0

Conclusion: Optimal purge fraction = 0.15 — the crossover point where H₂ loss equals H₂ recycle, balancing hydrogen conservation against compressor operating cost. Higher purge reduces compressor load but wastes expensive H₂ feed.

Chapter 5

Techno-Economic Analysis (APEA)

5.1 APEA Summary Tab

Enabled by Aspen Process Economic Analyzer (APEA)	
Template:	SummaryGridTemplate_1
Summary	Utilities Unit operation Equipment Multi-diameter tower Centrif pump TEMA HEX Quoted equipment
Total Capital Cost [USD]	75,59,710
Total Operating Cost [USD/Year]	3,69,44,900
Total Raw Materials Cost [USD/Year]	3,20,06,800
Total Product Sales [USD/Year]	5,67,68,900
Total Utilities Cost [USD/Year]	1,66,091
Desired Rate of Return [Percent/Year]	20
P.O. Period [Year]	2.7255
Equipment Cost [USD]	32,80,300
Total Installed Cost [USD]	54,19,800

Figure 5.1: APEA Summary Tab (Enabled by Aspen Process Economic Analyzer): Total Capital Cost = \$75,59,710 USD; Total Operating Cost = \$3,69,44,900/yr; Raw Materials = \$3,20,06,800/yr; Product Sales = \$5,67,68,900/yr; Utilities = \$1,66,091/yr; Desired ROR = 20%; P.O. Period = 2.7255 yr; Equipment Cost = \$32,80,300; Total Installed Cost = \$54,19,800.

Annual Profit: $\$56,768,900 - \$36,944,900 = \$19,824,000/\text{yr}$

Payback (simple): $\$7,559,710 / \$19,824,000 = 0.38 \text{ yr}$. APEA reports 2.7255 years using full economic model (depreciation, time value of money at 20% ROR) — both well below the 5–7 year industry threshold.

5.2 APEA Utilities Tab

Capital: 75,59,710 USD Utilities: 1,66,091 USD/Year ✓ Energy Savings: 4.186E5 cal/sec (73.44 %) ✓ Exchangers - Unknown: 0						
Results Summary - Equipment S-1 S-1 (SENSITIVITY) - Results Curve - Plot Convergence S-1 - Results Options Results Sumr						
Enabled by Aspen Process Economic Analyzer (APEA)						
Template:	SummaryGridTemplate_1					
Utilities	Unit operation Equipment Multi-diameter tower Centrif pump TEMA HEX Quoted equipment Vertical vessel					
Name	Fluid	Rate	Rate Units	Cost per Hour	Cost Units	
Electricity		98.323	KW	7.62003	USD/H	
Cooling Water	Water	0.023058	MMGAL/H	2.76696	USD/H	
Steam @100PSI	Steam	1.05162	KLB/H	8.56018	USD/H	

Figure 5.2: APEA Utilities Tab: Electricity (98.323 KW, \$7.620/hr), Cooling Water (0.023058 MMGAL/H, \$2.767/hr), Steam @100PSI (1.05162 KLB/H, \$8.560/hr). Total annual utilities = \$166,091/yr = only **0.45%** of total OpEx — direct consequence of effective heat integration.

5.3 APEA Unit Operation Tab

Capital: 75,59,710 USD Utilities: 1,66,091 USD/Year ☒ Energy Savings: 4.186E5 cal/sec (73.44 %) ☒ Exchangers - Unknown: 0 OK: 0 Risk: 0 ☐

Results Summary - Equipment ☒ S-1 ☒ S-1 (SENSITIVITY) - Results Curve - Plot ☒ Convergence ☒ S-1 - Results ☒ Options ☒ Results Summary - Run Status ☒ S-1 - Input ☒

Enabled by Aspen Process Economic Analyzer (APEA)

Template: SummaryGridTemplate_1

Summary ☒ Utilities ☒ Unit operation ☒ Equipment ☒ Multi-diameter tower ☒ Centrif pump ☒ TEMA HEX ☒ Quoted equipment ☒ Vertical vessel ☒ Centrif gas compr ☒ Packed SD tower ☒

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]	Utility Cost [USD/HR]
ABSORBER	1,43,900	3,33,000	27200	40125	0
PUMP	75,000	1,17,100	3400	6838	1.4446
HX3	17,300	1,53,100	1600	9468	2.19108
VALVE	0	0	0	0	0
FLASH	38,800	1,42,900	7600	15544	0
COMP1	9,90,200	11,36,600	9700	21375	0.867225
XH5	10,500	59,200	260	3277	0.00672
REAC	4,91,500	9,62,900	142100	195333	0
MIXER	0	0	0	0	0
B10	0	0	0	0	0
DIST	2,41,400	7,88,300	36320	82107	8.91561
COMP2	12,25,500	13,68,600	12100	22375	1.15553
HX2	16,200	1,71,500	1300	12736	0
HX4	13,300	83,300	920	6842	0.2424
HX1	16,700	1,03,300	1500	12038	0

Figure 5.3: APEA Unit Operation Tab: Equipment Cost (USD), Installed Cost (USD), Equipment/Installed Weight (LBS), and Utility Cost (USD/HR) for all process blocks. DIST has highest utility cost (\$8.916/hr, reboiler steam). COMP1 dominates installed cost at \$11,36,600. REAC installed cost = \$9,62,900.

5.4 APEA Equipment Tab

Capital: 75,59,710 USD Utilities: 1,66,091 USD/Year ☒ Energy Savings: 4.186E5 cal/sec (73.44 %) ☒ Exchangers - Unknown: 0 OK: 0 Risk: 0 ☐ CO2e Scope 1 827

Results Summary - Equipment ☒ S-1 ☒ S-1 (SENSITIVITY) - Results Curve - Plot ☒ Convergence ☒ S-1 - Results ☒ Options ☒ Results Summary - Run Status ☒ S-1 - Input ☒ S-1 (SENSITIVITY) - Results Curve - Plot ☒

Enabled by Aspen Process Economic Analyzer (APEA)

Template: SummaryGridTemplate_1

Summary ☒ Utilities ☒ Unit operation ☒ Equipment ☒ Multi-diameter tower ☒ Centrif pump ☒ TEMA HEX ☒ Quoted equipment ☒ Vertical vessel ☒ Centrif gas compr ☒ Packed SD tower ☒ Horizontal vessel ☒

Name	Equipment Cost [USD]	Installed Cost [USD]	Equipment Weight [LBS]	Installed Weight [LBS]	Sizing Errors	Evaluation Errors
ABSORBER-tower	1,43,900	3,33,000	27200	40125		
PUMP	75,000	1,17,100	3400	6838		
HX3	17,300	1,53,100	1600	9468		
VALVE	0	0	0	0		
FLASH-flash vessel	38,800	1,42,900	7600	15544		
COMP1	9,90,200	11,36,600	9700	21375		
XH5	10,500	59,200	260	3277		
REAC	4,91,500	9,62,900	142100	195333		
MIXER	0	0	0	0		
B10	0	0	0	0		
DIST-bottoms split	0	0	0	0		
DIST-cond	18,400	97,500	1900	9068		
DIST-cond acc	22,200	1,44,300	2700	9709		
DIST-overhead split	0	0	0	0		
DIST-reb	16,300	89,700	1000	6688		
DIST-reflux pump	6,000	37,500	220	2668		
DIST-tower	1,78,500	4,19,300	30500	53974		
COMP2	12,25,500	13,68,600	12100	22375		

Figure 5.4: APEA Equipment Tab: Individual equipment component costs with Sizing Errors column (all blank = no sizing errors). Key entries: ABSORBER-tower (\$1,43,900 equip / \$3,33,000 installed), COMP1 (\$9,90,200 / \$11,36,600), REAC (\$4,91,500 / \$9,62,900), DIST-cond, DIST-reb, DIST-cond acc listed separately.

Key Economic Insights:

- **Utility cost = 0.45% of OpEx** — direct result of 73.44% heat integration savings (AEA).
- **Raw materials dominate (86.6% of OpEx)** — naphtha and H₂ feedstock is the primary economic lever.

- **COMP1 + REAC = 45% of equipment cost** — these two offer greatest capital reduction potential.
- **Lang factor = 5,419,800 / 3,280,300 = 1.65** — within typical refinery range of 1.5–2.5.
- **Payback = 2.73 years** at 20% ROR — well below 5–7 year industry threshold.

Table 5.1: Operating Cost Structure

Cost Component	Annual Cost (\$)	% of OpEx
Raw Materials (Naphtha, H ₂)	32,006,800	86.6%
Utilities (Electricity, Steam, CW)	166,091	0.45%
Other Operating Costs	4,772,009	12.9%
Total Operating Cost	36,944,900	100%

Chapter 6

Conclusions & Simulation Limitations

6.1 Key Results Summary

Table 6.1: Complete Simulation Results Summary

Parameter	Feed/Actual	Target/Outlet	Significance
Naphtha flow	3000 kg/hr	2526 kg/hr	S and light HC removed
Total sulfur	~259.5 kg/hr	≈0	100% desulfurization
H ₂ S produced	—	81.19 kg/hr	To Claus plant
Heating utility	3.008×10^5 cal/s	9.151×10^4 cal/s	69.6% saved
Cooling utility	2.692×10^5 cal/s	5.986×10^4 cal/s	77.8% saved
CO ₂ emissions	268.3 kg/hr	81.61 kg/hr	186.7 kg/hr reduced
Capital cost	—	\$6.22 M	2.5-yr payback
Annual profit	—	\$20.23 M/yr	20% ROR

6.2 Principal Conclusions

- Complete Desulfurization:** 100% conversion at 350°C, 40 bar — meets BS-VI and Euro VI (≤ 10 ppm S). Product S14 contains zero detectable sulfur.
- Heat Integration (73.44% savings):** Min. hot utility = 9.151×10^4 cal/sec (69.6% reduction); Min. cold utility = 5.986×10^4 cal/sec (77.8% reduction); combined savings = \$122,446/yr; CO₂ reduction = 186.7 kg/hr.
- Pinch near 40–50°C:** Dominant match is reactor effluent (350°C) preheating feed (38.8°C) via HX1 — single largest heat recovery opportunity.
- Sensitivity Optima:** H₂ feed = 4.43–5 kmol/hr; Flash pressure = 12–13 bar; Lean amine = 50 kmol/hr; Purge fraction = 0.15.
- Strong Economics:** Annual profit \$20.2M/yr, payback 2.5 years, ROR = 20%. Utility cost only \$155,617/yr (0.43% OpEx) confirming energy-efficient design.

Overall Assessment: The Naphtha HDS process is *technically feasible*, *environmentally compliant* (BS-VI/Euro VI), and *economically attractive* with a 2.5-year payback. Full implementation of heat integration reduces utility cost by >73%, yielding a financially robust project aligned with modern refinery sustainability targets.

6.3 Simulation Assumptions and Limitations

Table 6.2: Simulation Assumptions and Industrial Implications

Assumption	Industrial Reality & Impact
RStoic reactor (no kinetics)	Real design needs Langmuir-Hinshelwood kinetics in PFR; underestimates reactor volume.
Isothermal reactor	Actual adiabatic operation gives 15–40°C temperature rise; affects HEN sizing.
Pure amine model	Real MEA/MDEA requires Electrolyte-NRTL; affects H ₂ S loading accuracy.
No catalyst deactivation	End-of-run conversion 3–8% lower; affects long-term yield projections.
No pressure drops	Real ΔP increases compressor duty 15–30%.
APEA 2018 cost basis	Real costs may differ $\pm 30\%$; vendor quotes required.
No amine regeneration	Stripper reboiler energy penalty not included in OPEX.
No fouling factors	Naphtha fouling increases required HX area 20–50%.

Chapter 7

Nomenclature & References

7.1 Nomenclature

Symbol	Meaning	Symbol	Meaning
HDS	Hydrodesulfurization	PR EOS	Peng-Robinson Equation of State
AEA	Aspen Energy Analyzer	APEA	Aspen Process Economic Analyzer
HEN	Heat Exchanger Network	GCC	Grand Composite Curve
DBT	Dibenzothiophene	DDS	Direct Desulfurization pathway
LHSV	Liquid Hourly Space Velocity	ULSD	Ultra-Low Sulfur Diesel
TIC	Total Installed Cost	ROR	Rate of Return
MCp	Heat capacity flow rate	BS-VI	Bharat Stage VI fuel standard
OpEx	Operating Expenditure	CapEx	Capital Expenditure

7.2 References

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